



Test Report issued under the responsibility of:



**TEST REPORT
IEC 62133-2**

**Secondary cells and batteries containing alkaline or other non-acid electrolytes – Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications –
Part 2: Lithium systems**

Report Number..... : 50231670 002

Date of issue..... : 2022-07-12

Total number of pages : 6 pages

**Name of Testing Laboratory
preparing the Report : Guangzhou MCM Certification & Testing Co., Ltd.**

Applicant's name : Xinxiang Hongli Supply Source Technology Co., Ltd.

**Address..... : West Section of Xinglong Road, Xinxiang Economic
Development Zone, Xinxiang City, Henan, P.R. China**

Test specification:

Standard : IEC 62133-2:2017, IEC 62133-2:2017/AMD1:2021

Test procedure : CB Scheme

Non-standard test method : N/A

TRF template used..... : IECEE OD-2020-F1:2021, Ed.1.4

Test Report Form No. : IEC62133_2B

Test Report Form(s) Originator : DEKRA Certification B.V.

Master TRF : Dated 2021-08-31

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General disclaimer:

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Test item description	Cylindrical Lithium-ion Rechargeable Cell	
Trade Mark(s)	N/A	
Manufacturer	Same as applicant	
Model/Type reference	18650-1200mAh	
Ratings	DC 3.7V, 1200mAh, 4.44Wh	
Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):		
☒ CB Testing Laboratory:	Guangzhou MCM Certification & Testing Co., Ltd.	
Testing location/ address	Room 101 to 116 & 216, Building 2 (Office Building and Workshop) No. 45 Zhong Er Section of Shiguang Road, Zhongcun Street, Panyu District, Guangzhou City, Guangdong Province, China	
Tested by (name, function, signature)	Zhipeng Liu (Engineer)	<i>ZhiPeng Liu</i>
Approved by (name, function, signature) ..	Liang Hongcheng (Reviewer)	<i>LiangHongcheng</i>
☐ Testing procedure: CTF Stage 1:		
Testing location/ address		
Tested by (name, function, signature)		
Approved by (name, function, signature) ..		
☐ Testing procedure: CTF Stage 2:		
Testing location/ address		
Tested by (name + signature)		
Witnessed by (name, function, signature) ..		
Approved by (name, function, signature) ..		
☐ Testing procedure: CTF Stage 3:		
☐ Testing procedure: CTF Stage 4:		
Testing location/ address		
Tested by (name, function, signature)		
Witnessed by (name, function, signature) ..		
Approved by (name, function, signature) ..		
Supervised by (name, function, signature) :		

List of Attachments (including a total number of pages in each attachment):	
-Attachment 1: National Difference (3 pages).	
Summary of testing:	
Tests performed (name of test and test clause): N/A	Testing location: N/A
Summary of compliance with National Differences (List of countries addressed): KR KR=Republic of Korea	
☒ The product fulfils the requirements of <u>EN62133-2: 2017+A1</u>	
Use of uncertainty of measurement for decisions on conformity (decision rule) : ☒ No decision rule is specified by the IEC standard, when comparing the measurement result with the applicable limit according to the specification in that standard. The decisions on conformity are made without applying the measurement uncertainty ("simple acceptance" decision rule, previously known as "accuracy method"). ☐ Other: N/A (to be specified, for example when required by the standard or client, or if national accreditation requirements apply)	
Information on uncertainty of measurement: The uncertainties of measurement are calculated by the laboratory based on application of criteria given by OD-5014 for test equipment and application of test methods, decision sheets and operational procedures of IECEE. IEC Guide 115 provides guidance on the application of measurement uncertainty principles and applying the decision rule when reporting test results within IECEE scheme, noting that the reporting of the measurement uncertainty for measurements is not necessary unless required by the test standard or customer. Calculations leading to the reported values are on file with the NCB and testing laboratory that conducted the testing.	
Copy of marking plate: The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks. See previous report 50231670 001.	

Test item particulars.....:	
Classification of installation and use.....:	To be defined in final product
Supply Connection	The positive and the negative
Recommend charging method declared by the manufacturer	Charging the cell with 240mA constant current until 4.2V, then constant voltage until charge current reduces to 24mA at ambient 20°C±5°C.
Discharge current (0,2 It A)	240mA
Specified final voltage.....:	3.0V
Upper limit charging voltage per cell.....:	4.2V
Maximum charging current	600mA
Charging temperature upper limit	45°C
Charging temperature lower limit.....:	0°C
Polymer cell electrolyte type.....:	☐ gel polymer ☐ solid polymer ☒ N/A
Possible test case verdicts:	
- test case does not apply to the test object.....: N/A	
- test object does meet the requirement.....: P (Pass)	
- test object does not meet the requirement.....: F (Fail)	
Testing.....:	
Date of receipt of test item	N/A
Date (s) of performance of tests	N/A
General remarks:	
"(See Enclosure #)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report.	
Throughout this report a ☐ comma / ☒ point is used as the decimal separator.	
Manufacturer's Declaration per sub-clause 4.2.5 of IEC60068-2-1:	
The application for obtaining a CB Test Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided:	☐ Yes ☒ Not applicable
When differences exist; they shall be identified in the General product information section.	
Name and address of factory (ies): Same as applicant	

General product information and other remarks:

This test report shall be read in conjunction with the previous report 50231670 001

Description of change(s):

1. Changed the name of applicant & manufacturer & factory from "Xinxiang hongli supply source technology co.,ltd" to "**Xinxiang Hongli Supply Source Technology Co., Ltd.**" see page 1, page 2, page 4, page 6 for details.
2. Changed the address of applicant & manufacturer & factory from "Dayang Village South, zhaipo Town, Xinxiang 453700, P. R. China" to "**West Section of Xinglong Road, Xinxiang Economic Development Zone, Xinxiang City, Henan, P.R. China**", see page 1, page 2, page 4 for details.
3. Upgraded test standard, from "IEC 62133-2:2017" to "**IEC 62133-2:2017, IEC 62133-2:2017/AMD1: 2021**", see page 1, page 6 for details.
4. Upgraded the fulfilled standard from "EN 62133-2:2017" to "**EN 62133-2:2017+A1**", see page 3 for details.
5. Added the National Differences (Republic of Korea), see page 3, Attachment 1 for details.

For the above described change(s) the following was considered to be necessary:

Change	Testing	Comments	Result
1, 2	N/A	1. Only different in writing and the location does not change. 2. No safety impact, no further testing considered as necessary.	P
3, 4, 5	N/A	No safety impact, no further testing considered as necessary.	P

History of amendments and modifications:

Ref. No. 50231670 001, dated 2019-04-24 (original test report)

Ref. No. 50231670 002, dated 2022-07-12 (1st amendment)

TABLE: Critical components information					P
Object / part No.	Manufacturer / trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾
Cell	Xinxiang Hongli Supply Source Technology Co., Ltd.	18650-1200mAh	DC 3.7V, 1200mAh	IEC 62133-2:2017, IEC 62133-2:2017/AMD1:2021	Tested with appliance
Supplementary information:					
¹⁾ Provided evidence ensures the agreed level of compliance. See OD-CB2039.					

-- End of Report --

Attachment 1

Report No.: 50231670 002

IEC62133_2B ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
ATTACHMENT TO TEST REPORT IEC 62133-2 (Republic of Korea) NATIONAL DIFFERENCES (Secondary cells and batteries containing alkaline or other non-acid electrolytes - Safety requirements for portable sealed secondary lithium cells, and for batteries made from them, for use in portable applications - Part 2: Lithium systems)			
Differences according to : National standard KC62133-2(2020-07)			
TRF template used: : IECEE OD-2020-F3, Ed. 1.1			
Attachment Form No. : KR_ND_IEC62133_2B			
Attachment Originator : KTR			
Master Attachment : Dated 2022-05-27			
Copyright © 2020 IEC System for Conformity Testing and Certification of Electrical Equipment (IECEE), Geneva, Switzerland. All rights reserved.			
	National Differences		
7.3.6	Over-charging of battery		N/A
(Revision)	[Add the bolded text] b) Test The test shall be carried out in an ambient temperature of 20 °C ± 5 °C. Each test battery shall be discharged at a constant current of 0,2 It A, to a final discharge voltage specified by the manufacturer. Sample batteries shall then be charged at a constant current of 2,0 It A, using a supply voltage which is: • 1,4 times the upper limit charging voltage presented in Table A.1 (but not to exceed 6,0 V) for single cell/cell block batteries or • 1,2 times the upper limit charging voltage presented in Table A.1 per cell for series connected multi-cell batteries, and • sufficient to maintain a current of 2,0 It A throughout the duration of the test or until the supply voltage is reached. <u>• In case the charging voltage specified by the manufacturer is higher than the overcharge test voltage, the maximum charging voltage specified by manufacturer should be applied with 2.0 ItA,</u> <u>(e.g., quick charging power bank, etc.)</u>	Cell only	N/A

IEC62133_2B ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	[Replace to the following statement] c) Acceptance criteria Filling beyond the manufacturer's specified limits should not result in ignition or explosion		N/A
Annex G	Definition for shape and materials of outer case for cell		—
(Addition)	G.1 General Annex G provides definitions for shape and materials of outer case for cell G.2 Shape of outer case for cell G 2.1 Cylindrical cell Cell with a cylindrical shape in which the overall height is equal to or greater than diameter. G 2.2 Prismatic cell Cell having the shape of a parallelepiped whose faces are rectangular G.3 Materials of outer case for cell G.3.1 Soft case Non-metallic outer case or container for cell G.3.2 Hard case Metallic outer case or container for cell.	(Shape of outer cases) ☒ Cylindrical ☐ Prismatic (Materials of outer cases) ☒ Hard ☐ Soft	—
Annex H	Calculation method of the volumetric energy density for cell		—
(Addition)	Annex H provide a calculation method of the volumetric energy density for cell in use of smart phone, tablet, notebook. H.1 General Unless otherwise stated in the Annex E, the dimensions for calculation are based on these for cell before shipment and the volumetric energy density shall be calculated with a maximum values specified by manufacturer. If the specification for cell can't be provided a dimension for calculation, the manufacturer's other documentation shall be provided to demonstrate compliance for its calculation.	257.72Wh/ L	—

IEC62133_2B ATTACHMENT			
Clause	Requirement + Test	Result - Remark	Verdict
	H.2 Calculation Method L : Length (max.) of cell (including terrace) W : Width (max.) of cell T : Thickness (max.) when shipping charge (For reference, Please Exclude the dimension of any tape that is attached to cell) $\text{Volumetric energy density (Wh/L)} = \frac{\text{Nominal voltage (V)} \times \text{Rated capacity (Ah)}}{\text{Length (L)} \times \text{Width (W)} \times \text{Thickness (T)}}$ [H.1 – Prismatic cell using soft case] L : Length (max.) of cell W : Width (max.) of cell T : Thickness when shipping charge (For reference, Please Exclude the dimension of any tape that is attached to cell) $\text{Volumetric energy density (Wh/L)} = \frac{\text{Nominal voltage (V)} \times \text{Rated capacity (Ah)}}{\text{Length (L)} \times \text{Width (W)} \times \text{Thickness (T)}}$ [H.2 – Prismatic cell using hard case] D : Diameter (max.) of cell L : Length (max.) of cell (According to shape of cell at shipping, The dimension of tube for cell may be included in overall dimension of cell) $\text{Volumetric energy density (Wh/L)} = \frac{\text{Nominal voltage (V)} \times \text{Rated capacity (Ah)}}{3.14159 \times \frac{\text{Diameter (D)}^2}{4} \times \text{Length(L)}}$ [H.3 – Cylindrical cell using hard case]		